

# Launching VPC Resources

Melvin green

**Create VPC** [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

**VPC settings**

**Resources to create** [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

**Name tag auto-generation** [Info](#)

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

**IPv4 CIDR block** [Info](#)

Determine the starting IP and the size of your VPC using CIDR notation.

CIDR block size must be between /16 and /28.

**IPv6 CIDR block** [Info](#)

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

**Tenancy** [Info](#)

**Encryption settings - optional**

**VPC encryption control (\$)** [Info](#)

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply [L2](#)

☒ None ☐ Monitor mode ☐ Enforce mode

**Preview**

**VPC** [Show details](#)

Your AWS virtual network

Green VPC-vpc

**Subnets (6)**

Subnets within this VPC

**us-east-1a**

- Green VPC-subnet-public1-us-east-
- Green VPC-subnet-private1-us-
- Green VPC-subnet-private3-us-

**us-east-1b**

- Green VPC-subnet-public2-us-east-
- Green VPC-subnet-private2-us-
- Green VPC-subnet-private4-us-

**Route tables (5)**

Route network traffic to resources

- Green VPC-rtb-public
- Green VPC-rtb-private1-us-east-
- Green VPC-rtb-private2-us-east-
- Green VPC-rtb-private3-us-east-
- Green VPC-rtb-private4-us-east-

Activate Windows  
Go to Settings to activate

## Introducing Today's Project!

### What is Amazon VPC?

What is Amazon VPC, and why is it useful?9:22 PM Claude responded: Amazon VPC (Virtual Private Cloud) is a service that lets you launch AWS resources in a logically isolated virtual network that you define. Amazon VPC (Virtual Private Cloud) is a service that lets you launch

AWS resources in a logically isolated virtual network that you define. Think of it as your own private section of the AWS cloud.

## How I used Amazon VPC in this project

"I used Amazon VPC today in this project by creating a VPC with public and private components, including subnets, routing tables, an internet gateway, ACLs, and security groups. I also launched both public and private EC2 instances within the VPC, configured key pairs for secure SSH access, and used the 'VPC and more' option to streamline the setup process."

## One thing I didn't expect in this project was...

The one thing I did not expect from the project was how quickly I could create a whole VPC infrastructure using the 'VPC and more' option. Now, having the ability to create a large number of VPCs in a very short amount of time will be very valuable in an enterprise setting. The benefit of this is that in an enterprise setting, where multiple VPCs need to be deployed quickly and efficiently, this skill will save significant time and reduce the risk of configuration errors when setting up VPC components such as subnets, routing tables, internet gateways, ACLs, and security groups.

## This project took me...

This project took me a little over an hour, including demo time and learning the new material.

# Setting Up Direct VM Access

Directly accessing a virtual machine means "logging into" the EC2 instance (instead of just managing it at a higher level with the AWS Management Console). This includes operations like installing software and changing my EC2 Instance's configurations.

## SSH is a key method for directly accessing a VM

SSH means secure shell, and it is a protocol and a traffic type. It is the protocol that matches key pairs, and enables direct VM access, and once a connection is set up, it is a traffic type that encrypts communication data being transferred.

## To enable direct access, I set up key pairs

Key pairs are tools that help engineers authenticate when requesting direct access to a virtual machine, e.g., an EC2 instance. Key pairs work by having two keys — a private key for the VM, and a matching public key for the user trying to get access.

A private key's file format means the file type in which my key is stored. My private key's file format was .pem, which is a widely accepted file format that most servers will be able to read/use.

# Launching a public server

Start your response with 'I had to change my EC2 instance's networking settings by changing the VPC and the subnet my EC2 instance was going to be launched in! I updated both my VPC and my public subnet, my EC2 instance was going to be launched in! I updated both my VPC and my Public subnet. I also used my existing public security group.

|                                                                                          |                                                                               |                                             |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------|
| i-0859834c178831fb1 (Mels Public Server)                                                 |                                                                               |                                             |
| Details   Status and alarms   Monitoring   Security   <b>Networking</b>   Storage   Tags |                                                                               |                                             |
| VPC ID<br>vpc-011e1be523fe06845 (Mels VPC) <a href="#">t</a>                             | Subnet ID<br>subnet-0777496623f7b1d70 (Mels Public Subnet ) <a href="#">t</a> | Availability zone<br>us-east-1a             |
| Availability zone ID<br>use1-az1                                                         | Outpost ID<br>-                                                               |                                             |
| ▼ IP addresses <a href="#">Info</a>                                                      |                                                                               |                                             |
| Public IPv4 address<br>3.227.229.149   <a href="#">open address t</a>                    | Private IPv4 addresses<br>10.0.0.183                                          | IPv6 addresses<br>-                         |
| Secondary private IPv4 addresses<br>-                                                    | Carrier IP addresses (ephemeral)<br>-                                         |                                             |
| ▼ Hostname and DNS <a href="#">Info</a>                                                  |                                                                               |                                             |
| Public DNS<br>-                                                                          | Private IP DNS name (IPv4 only)<br>ip-10-0-0-183.ec2.internal                 | IPv4-only IP based name: A record only<br>- |
| Dualstack - IP based name: A and AAAA record<br>-                                        | IPv6-only - IP based name: AAAA record only<br>-                              | Public hostname type<br>-                   |

# Launching a private server

My private server has its own dedicated security group because the public security group allows all HTTP traffic, which would leave our private server much more vulnerable to security attacks and risks.

My private server's security group's source is Mel's private security group, which means only SSH traffic coming from resources associated with that security group would be allowed.

The screenshot displays the 'Configure Security' step in the AWS Management Console. It shows the configuration for a new security group named 'Mels private security group' within the 'Mels VPC' and 'Mels Private Subnet'. The 'Create security group' option is selected. The security group name is 'Mels private security group' and the description is 'launch-wizard-1 created 2026-05-28T00:01:56.280Z'. Under 'Inbound Security Group Rules', a rule is added for 'ssh' traffic on port 22, allowing access from the security group 'sg-0c307d5f1fee190f1'.

**VPC - required** | Info  
vpc-011e1be523fe06845 (Mels VPC)  
10.0.0.0/16

**Subnet** | Info  
subnet-0ceac2a6d046bda3c Mels Private Subnet  
VPC: vpc-011e1be523fe06845 Owner: 157975550086 Availability Zone: us-east-1b (use1-az2)  
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

**Auto-assign public IP** | Info  
Enable

**Firewall (security groups)** | Info  
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.  
☒ Create security group ☐ Select existing security group

**Security group name - required**  
Mels private security group  
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and \_-./!@#%&\*(){}\$%.

**Description - required** | Info  
launch-wizard-1 created 2026-05-28T00:01:56.280Z

**Inbound Security Group Rules**  
▼ Security group rule 1 (TCP, 22, sg-0c307d5f1fee190f1) Remove

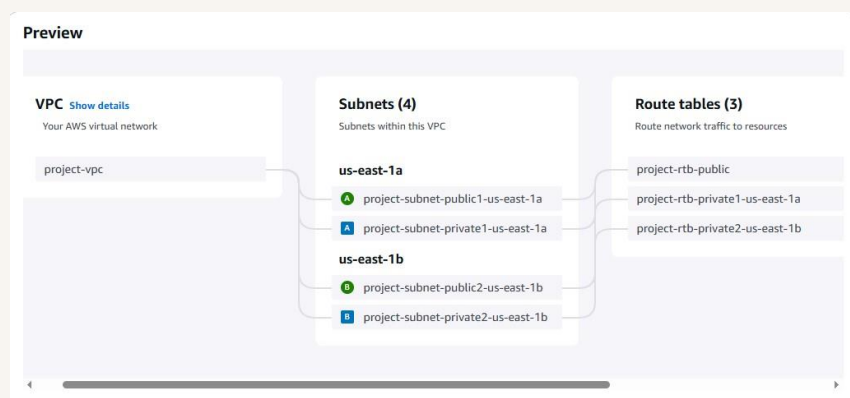
| Type | Protocol | Port range | Source type | Source               | Description - optional     |
|------|----------|------------|-------------|----------------------|----------------------------|
| ssh  | TCP      | 22         | Custom      | sg-0c307d5f1fee190f1 | e.g. SSH for admin desktop |

# Speeding up VPC creation

I used an alternative way to set up an Amazon VPC. This time, I used the 'VPC and more' option, which provided a VPC resource map for creating a VPC and all of its components, including security groups, route tables, and internet gateways.

A VPC resource map is a visual diagram that maps out my VPC's components and the relationships/connections between them. A resource map is an interactive, i.e., it will highlight the connections relevant to a resource I highlight/hover over.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because VPCs are already isolated from each other. Still, this is not the best practice if we need VPC peering.



# Speeding up VPC creation

## Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: either none or one in each availability zone. This is because it is best practice to improve redundancy and high availability by having at least one subnet per availability zone.

The setup page also offered to create NAT gateways, which are connector gateways that would let resources in my private subnet get access to the internet while still blocking off incoming traffic from the internet.

The screenshot displays the 'Create VPC' page in the AWS Management Console. The left sidebar contains the 'VPC settings' section, which includes options for 'Resources to create' (VPC only or VPC and more), 'Name tag auto-generation' (Auto-generate with tag 'Green VPC'), 'IPv4 CIDR block' (10.0.0.0/16), 'IPv6 CIDR block' (No IPv6 CIDR block), 'Tenancy' (Default), and 'Encryption settings - optional' (VPC encryption control set to None). The main area shows a 'Preview' of the VPC resource map. This map illustrates the 'Green VPC-vpc' connected to six subnets: one public subnet in us-east-1a and five private subnets (one in us-east-1a and four in us-east-1b). These subnets are then connected to five route tables: one public route table and four private route tables, each associated with a specific subnet.

